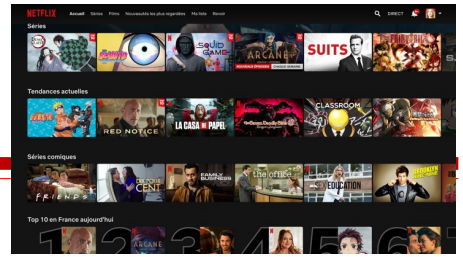
IDS = Introduction to Distributed Systems
2024 - 2025

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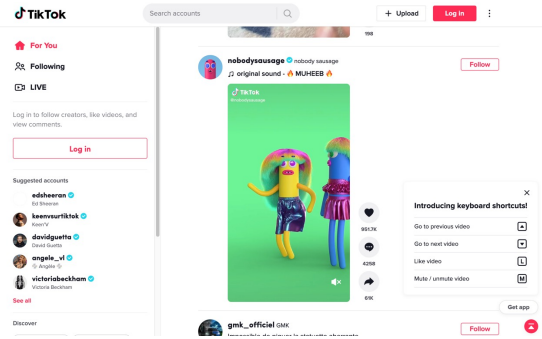
Université Grenoble Alpes
LIG Laboratory

Distributed systems are everywhere

- **mail, discord, mattermost...**
- **gaming**
- **web**
- **Deezer, Apple Music, ...**
- **Netflix, ...**
- **ChatGPT, ...**
- **...**
- **datacenter**
- **HPC cluster**
- **...**
- **cloud**
- **...**

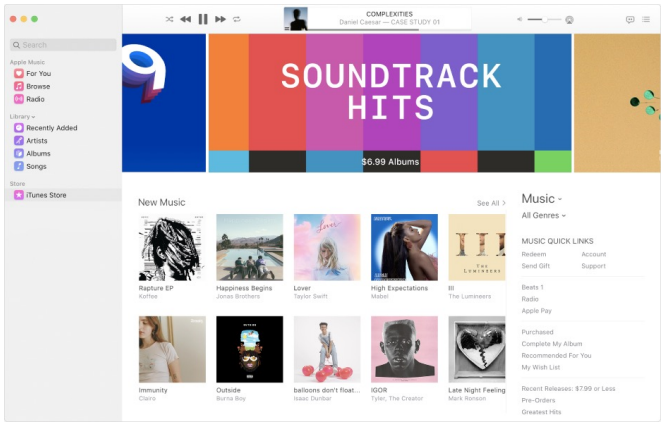








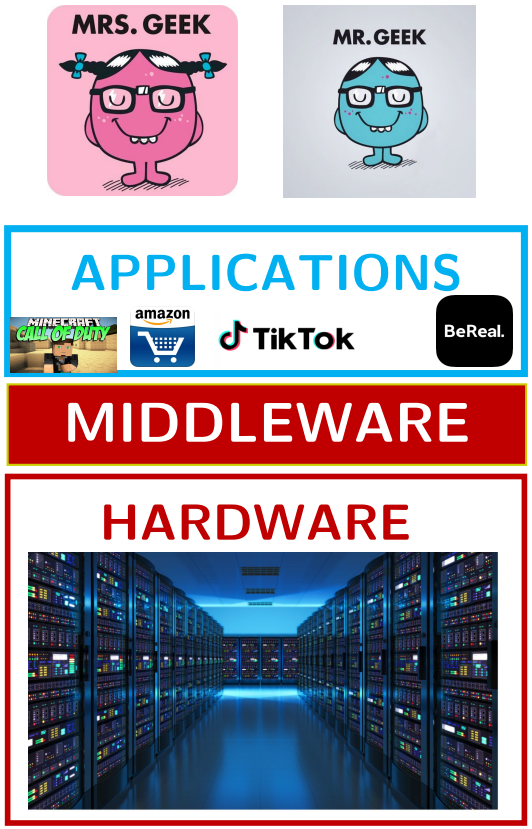






Goals: Be aware, control the usage, the functionality

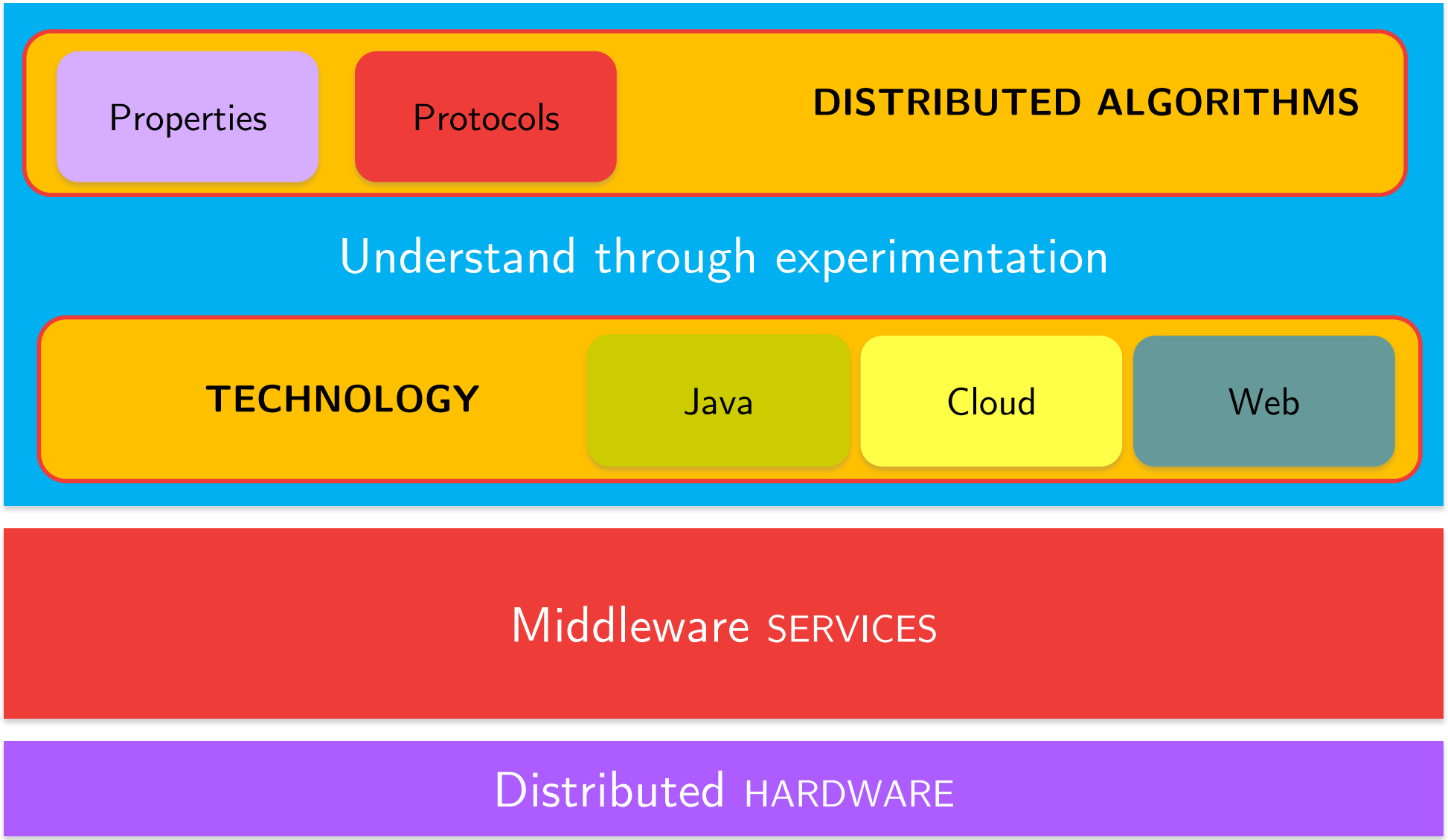
- **What is it about ?**
- **Is it difficult, I mean really ?**
- **What algorithms = what is the functionality, is it correct ?**
- **What technology = how is it implemented ?**



Get to know the foundations

- architecture
- client-server
- protocol
- coordination, correction
- service, interface
- middleware
- tier
- patterns
- ...

Ambitions


Properties
Protocols
DISTRIBUTED ALGORITHMS
Understand through experimentation
TECHNOLOGY
Java
Cloud
Web
Middleware SERVICES
Distributed HARDWARE

Organization

- **Come if you like coding, experimenting and learning**
- **Lectures + labs**
  - Labs 50%
  - Final exam 50%
- **The exact organization will depend on the number of students**

  - Distributed chat
  - Distributed game
  - Distributed shared memory
  - Virtual network (overlay)